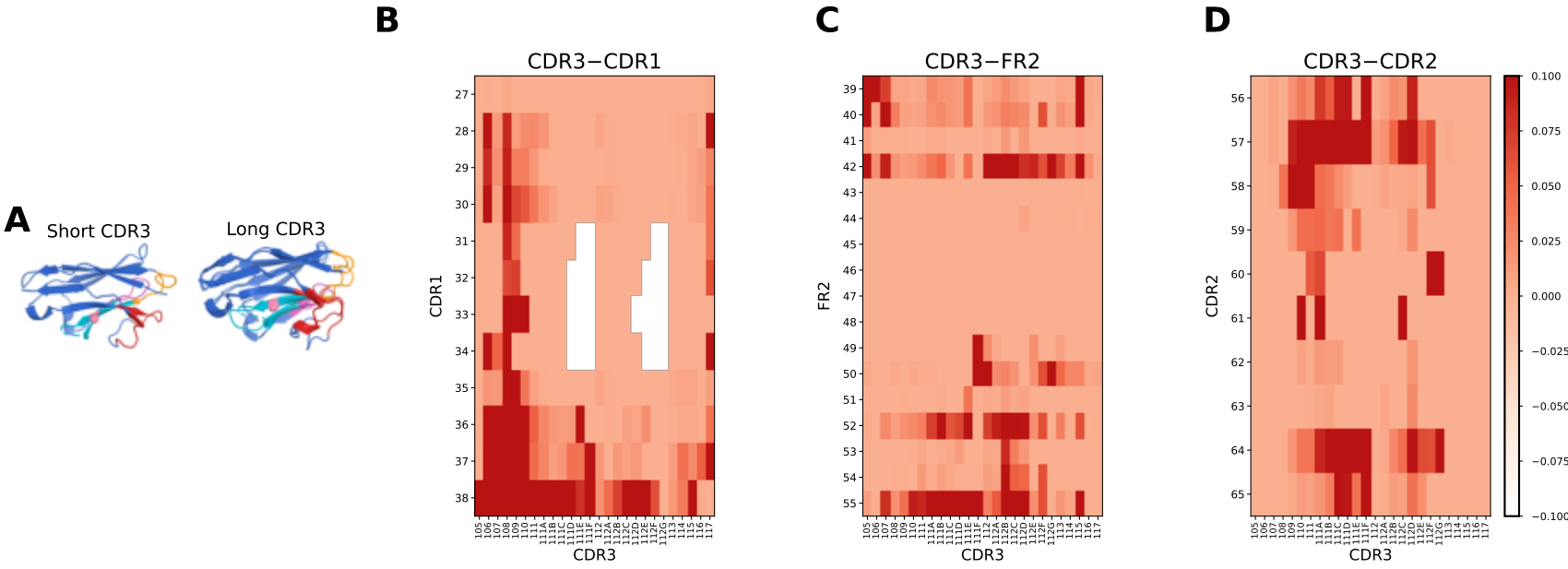


E

Group	CDR3 interactions with other regions				CDR3 second structure						
	CDR1	FR2	CDR2	FR2_CDR3_Dis	G	T	E	S	C	SASA	Surface hydrophob.
VHH	0.40 ****	0.60 ****	0.58 ****	-0.53 ****	0.37 ****	-0.20 ****	-0.44 ****	-0.06 ns	0.42 ****	0.48 ****	-0.27 ****
Conventional VH	0.42 ****	0.19 ****	0.43 ****	-0.21 ****	0.06 *	-0.16 ****	-0.09 ***	-0.18 ****	0.30 ****	0.77 ****	-0.11 ****

FIGURE 4B

CDR3 length-dependent structural organization (VHH). (A) Representative short-CDR3 (left) and long-CDR3 (right) VHH exemplars (Q25/Q75 CDR3 length, deterministic); CDR3=red, CDR1=orange, FR2=cyan, CDR2=magenta, IMGT 42=pink sphere. (B-D) CDR3-region interaction probability (≤4Å) with CDR1 (B), FR2 (C), and CDR2 (D) across the VHH cohort (N=533), shared color scale. (E) Pearson correlation between CDR3 length and structural/physicochemical features, VHH (N=533) vs conventional VH (N=1428-1429); cell color = correlation sign/magnitude, value = r, significance below = BH-FDR q (ns/*/**/***/**** = q≥0.05/<0.05/<0.01/<0.001/<0.0001). Solubility is intentionally excluded (not computed, per protocol); surface charge/LPP/LNP are omitted (PEP-Patch not reproduced in this toolchain, see Figure4B_metric_definition_status.csv). Formal CDR3-length×group interaction statistics (β3, CI, q) are retained in the source data and caption but not shown as an additional heatmap here.